
MA 102: Mathematics II

Course Code	MA 102
Course Name	Mathematics II
L-T-P-C	2 -1 -0- 3
Prerequisites	Consent of the faculty member
Intended for	B.Tech. First Year
Distribution	Core
Approval	1st Adhoc Senate (Apr 2011)

Course Contents

- **Integral Calculus**
Double and Triple Integrals, Change of Order of Integration, Change of Variables, Gamma, Beta functions, Dirichlet's Integral. Application (Evaluation of surface area, Volume, Centre of Gravity, Moment of Inertia). [Lectures]
- **Vector Calculus**
Differentiation of Vectors, Gradient, Divergence, Curl and their Physical meaning, Differential Operators and their identities. Line and Surface Integrals. Green's Theorem in a plane. Gauss Divergence Theorem and Stoke's theorem and their applications. [Lectures]
- **Laplace Transform**
Definition, Shifting Theorems, Transform of Derivatives, Differentiation and Integration. Differentiation and Integration of Transforms, Heaviside unit step and Dirac-Delta functions. Inverse Laplace Transforms, Solution of Ordinary Differential Equations in Mechanics, Electric Circuits and bending of Beams using Laplace Transforms.
- **Fourier Series**
Trigonometric Fourier Series. Half Range series, Harmonic Analysis.
- **Fourier Transforms**
Definition, Fourier sine and Cosine Transforms, Fourier Integral Formula and Applications

Text Books

1. E.Kreyszig, **Advanced Engineering Mathematics**, 9th Edition, John Wiley,
2. George B. Thomas, Maurice D. Weir, Joel Hass, Frank R. Giordano, **Thomas'Calculus**, 11th Edition, Pearson, 2004.
3. Lokenath Debnath, Dambaru Bhatta, **Integral Transforms And Their Applications**, 2nd Edition, Chapman & Hall/CRC (2006).

References

1. Ian N. Sneddon, **Fourier Transforms**, Dover Publications (2010).